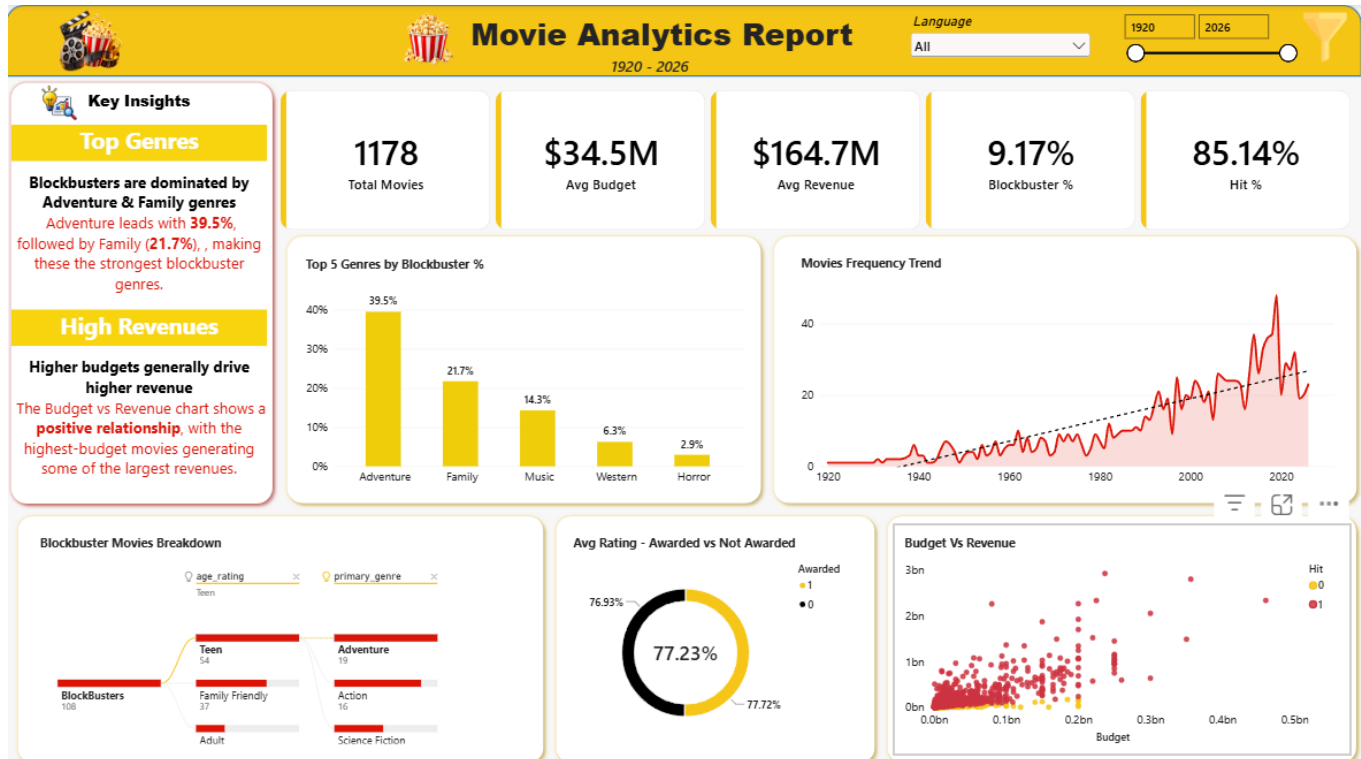


MOVIES ANALYTICS REPORT

End-to-End Data Science Project

Web Scrapping • Data Cleaning • EDA • Machine Learning • Power BI



Executive Summary

This project presents an end-to-end movie analytics workflow using publicly accessible movie information collected from The Movie Database (TMDB). The workflow covered two-layer web scraping, data cleaning and feature engineering, exploratory data analysis, classification modeling, and an interactive Power BI dashboard.

A total of 2,000 movie records were collected. After preprocessing, the dataset retained all 2,000 movies for general analytics, while 1,178 movies had both budget and revenue available and were therefore used for profitability-based analysis and the hit/flop classification model.

Project outcome

The final dashboard focuses on movie volume, financial performance, blockbuster share, hit rate, genre patterns, yearly trends, and relationships between budget and revenue.

Project at a Glance

Component	Implemented work
Data source	The Movie Database (TMDB)
Records scraped	2,000 movies
Scraping approach	Requests + BeautifulSoup; two-layer listing/detail-page extraction
Analytics dataset	2,000 cleaned movie records
Financially complete records	1,178 records with both budget and revenue
ML task	Binary classification: IsHit
Models tested	Logistic Regression, Decision Tree, Random Forest
Dashboard	Power BI Movie Analytics Report

1. Project Objective

The objective was to transform publicly available movie web data into a structured analytics dataset, identify meaningful patterns through EDA, build a machine-learning model to classify movie success, and communicate the results through an interactive Power BI dashboard.

- Collect 2,000 movie records through reproducible web extraction.
- Convert raw web data into a structured analytics-ready table.
- Explore movie trends across time, genres, ratings, and financial measures.
- Build a classification model for the profit-based IsHit target.
- Present the major findings through an interactive Power BI report.

2. Data Collection Method

Movie data was collected from TMDB's top-rated movie pages using Python. The scraping process used Requests to retrieve web pages and BeautifulSoup to parse the HTML.

Two-layer scraping approach

1. The first layer iterated through 100 pages of the TMDB top-rated movie listing and collected movie links.
2. The second layer requested each individual movie page and extracted the detailed fields.
3. The extracted fields were stored in Python lists and combined into a Pandas DataFrame.
4. The raw dataset was exported to a CSV file for subsequent analysis.

Fields collected in the raw dataset included:

- Title
- Year
- Certification
- Genres
- Runtime
- Director
- Awards
- Original language
- Budget
- Revenue
- Rating

Raw data output

The scraping stage produced a 2,000-row raw CSV dataset that served as the starting point for cleaning and preprocessing.

Code Screenshots -

Web Scraping Movies Data

```
In [ ]: # 1st layer
import requests
```

```
In [ ]: base_url = 'https://www.themoviedb.org/movie/top-rated'
```

```
In [ ]: from bs4 import BeautifulSoup
```

```
In [ ]: # We are getting 2000 movie links
movie_links = []
for page in range(1, 101):

    url = f"{base_url}?page={page}"

    response = requests.get(url)

    print(f"Scraping page {page}...")

    soup = BeautifulSoup(response.text, "html.parser")

    for j in soup.find_all('a', attrs={'class': "font-normal hover:underline
hover:underline-offset-2 hover:decoration-current hover:text-tmdb-light-
blue"}):
        s = 'https://www.themoviedb.org'+j.get('href')
        movie_links.append(s)
```

Same web scraping procedure for 2000 movies (Looping)

```
In [ ]: import time

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/151.0.0.0 Safari/537.36",
    "Accept":
"text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/web
p,*/*;q=0.8",
    "Accept-Language": "en-US,en;q=0.9",
    "Accept-Encoding": "gzip, deflate, br",
    "Connection": "keep-alive",
    "Upgrade-Insecure-Requests": "1"
}

for i, url in enumerate(movie_links):

    try:
        fly = requests.get(url, headers=headers, timeout=15)

        if fly.status_code != 200:
            print(f"Failed: {i+1} | Status: {fly.status_code}")
            continue

        data = BeautifulSoup(fly.text, 'html.parser')

        # ----- TITLE -----
        title_tag = data.find('h2')

        if title_tag and title_tag.find('a'):
            c1 = title_tag.find('a').get_text(strip=True)
        else:
            c1 = None
```

```

# ----- YEAR -----
if title_tag and title_tag.find('span'):
    c2 = title_tag.find('span').get_text(strip=True)
    c2 = c2.replace('(', '').replace(')', '')
else:
    c2 = None

years.append(c2)

# ----- CERTIFICATION -----

```

Jupyter Notebook — generated with runcell

```

tag = data.find('span', attrs={'class': 'certification'})

if tag:
    c3 = tag.get_text(strip=True)
else:
    c3 = None

certifications.append(c3)

# ----- GENRES -----
tag = data.find('span', attrs={'class': 'genres'})

if tag:
    c4 = tag.get_text(" ", strip=True)
else:
    c4 = None

genres.append(c4)

```

```
# ----- RUNTIME -----
tag = data.find('span', attrs={'class': 'runtime'})

if tag:
    c5 = tag.get_text(" ", strip=True)
else:
    c5 = None

runtimes.append(c5)

# ----- DIRECTOR -----
profile = data.find('li', attrs={'class': 'profile'})

if profile and profile.find('a'):
    c6 = profile.find('a').get_text(strip=True)
else:
    c6 = None

directors.append(c6)
```

```

# ----- FACTS -----
facts = data.find('section', attrs={'class': 'facts left_column'})

original_language = None
budget = None
revenue = None

if facts:

    for p in facts.find_all('p'):

        bdi = p.find('bdi')

        if not bdi:
            continue

        label = bdi.get_text(strip=True)
        value = p.get_text(" ", strip=True)

        value = value.replace(label, '', 1).strip()

        if label == 'Original Language':
            original_language = value

        elif label == 'Budget':
            budget = value.replace('$', '').replace(',', '')

        elif label == 'Revenue':
            revenue = value.replace('$', '').replace(',', '')

    original_languages.append(original_language)
    budgets.append(budget)
    revenues.append(revenue)

```

```

# ----- RATING -----
tag = data.find('div', attrs={'class': 'user_score_chart'})

if tag:
    c11 = tag.get('data-percent')
else:
    c11 = None

ratings.append(c11)

```

Jupyter Notebook — generated with runcell

```

print(f"{i+1}/{len(movie_links)} - {c1}")

# Small delay
time.sleep(0.5)

except Exception as e:

    print(f"ERROR: {i+1} | {url}")
    print(e)

# Keep lists aligned
titles.append(None)
years.append(None)
certifications.append(None)
genres.append(None)
runtimes.append(None)
directors.append(None)
awards.append(None)
original_languages.append(None)

```

Creating a Pandas Dataframe from the scraped data

```
In [ ]: import pandas as pd
df = pd.DataFrame({
    'title': titles,
    'year': years,
    'certification': certifications,
    'genres': genres,
    'runtime': runtimes,
    'director': directors,
    'awards': awards,
    'original_language': original_languages,
    'budget': budgets,
    'revenue': revenues,
    'rating': ratings
})
```

Converting the Dataframe to CSV file and Now, we have the Raw data file for the Analysis

```
In [ ]: df.to_csv('movies_data.csv')
```

```
In [ ]:
```

3. Data Cleaning & Preprocessing

The raw dataset contained missing values, mixed certification formats, text-based runtime values, non-numeric financial fields, and multiple genre formats. The preprocessing workflow converted these fields into structured variables suitable for analysis and modeling.

Missing values

Raw field	Missing values	Treatment
Certification	20	Filled with 'Unknown' before grouping
Director	2	Filled with 'Unknown'
Awards	1,442	Converted to binary 'awarded' indicator
Budget	724	Converted to numeric; '-' treated as missing
Revenue	550	Converted to numeric; '-' treated as missing

The cleaning notebook also identified zero exact duplicate rows. Sixteen duplicate-title occurrences were found, but the inspection showed that repeated titles could represent different movies with different years/directors, so title duplication alone was not treated as a duplicate-row rule.

Standardization and feature engineering

Transformation	Result
Certification	Grouped many rating systems into Family Friendly, Teen, Adult, and Unknown
Awards	Created binary 'awarded' feature: 1 if award information was present, otherwise 0
Genres	Standardized separators and extracted 'primary_genre'
Runtime	Converted strings such as hours/minutes into 'runtime_minutes'
Budget / Revenue	Converted text values to numeric; '-' became missing
Invalid small budget values	Budget values below 100 were set to missing
Profit	Created as revenue – budget
ROI	Created as (revenue – budget) / budget × 100
Genre count	Created 'genre_count' from the number of listed genres

Financial completeness

After cleaning, 1,276 movies had a recorded budget and 1,450 had a recorded revenue. There were 1,178 movies with both budget and revenue available. This 1,178-record subset was used for profitability-based analysis and the machine-learning classification task.

4. Exploratory Data Analysis

EDA was used to understand movie distributions, time trends, genre behavior, ratings, financial relationships, correlations, and outliers before modeling.

Summary statistics

Metric	Mean	Median	Min	Max
Rating	77.23	77	74	92
Runtime (minutes)	113.61	112	4	367
Budget (\$M)	32.64	12.00	0.000117	460.00
Revenue (\$M)	136.66	30.39	0.000006	2,923.71
Profit (\$M)	130.19	29.46	-185.60	2,686.71
ROI (%)	841.57	285.69	-100.00	67,332.90

Major EDA findings

- The collected movie sample contains 2,000 records spanning years from 1920 to 2026.
- The movie count trend increases strongly toward recent years, with the highest frequency visible around 2020 in the collected sample.
- Drama is the most frequent primary genre, with 682 movies, followed by Comedy and Animation with 233 each.
- Average rating is relatively concentrated, with an overall mean of 77.23 and a standard deviation of 2.81.
- Average revenue is much larger than median revenue, indicating a strongly right-skewed financial distribution.
- Revenue and budget show a strong positive correlation of approximately 0.77.
- Rating and revenue have only a weak positive correlation of approximately 0.14, indicating that higher ratings alone do not strongly explain commercial revenue.
- Average profit is strongly skewed; a small number of movies generate exceptionally large profits.

- Adventure has the highest average profit among the listed primary genres, followed by Science Fiction and Family.
- The boxplots show substantial financial and runtime outliers; these were treated as observations to investigate rather than automatically remove.

Key analytical message

Movie ratings are relatively concentrated, while financial outcomes vary dramatically. Budget is much more strongly associated with revenue than rating is, which makes financial scale an important part of the commercial-performance story.

5. Machine Learning Model

The machine-learning task was framed as a binary classification problem: predict the 'IsHit' target. IsHit was defined in the EDA workflow from profitability, where a movie with positive profit was assigned 1 and a non-positive profit was assigned 0.

Because IsHit is derived from profit, the model deliberately excluded profit, profit_in_mil, ROI, revenue, revenue_in_mil, and the IsHit target itself from the predictors. Budget was also excluded in its raw form while the engineered budget_in_mil feature was retained.

Modeling workflow

1. Restricted the modeling data to the 1,178 records with both budget and revenue available.
2. Separated predictors (X) from the IsHit target (y).
3. One-hot encoded primary_genre, original_language, and age_rating.
4. Used year, rating, runtime_minutes, genre_count, awarded, budget_in_mil and the encoded categorical variables as predictors.
5. Used an 80/20 stratified train-test split with random_state=42.
6. Standardized the numerical predictors.
7. Applied SMOTE to balance the training data.
8. Trained and evaluated Logistic Regression, Decision Tree, and Random Forest classifiers.

Model performance

Model	Accuracy	Precision*	Recall*	F1*
Logistic Regression	69.1%	88.1%	73.6%	80.2%
Decision Tree	73.3%	88.3%	79.1%	83.5%
Random Forest	76.3%	86.4%	85.6%	86.0%

*The precision, recall, and F1 values in this table are for the positive 'Hit' class, as reported by the model evaluation outputs.

Model interpretation

- Random Forest produced the highest overall accuracy at 76.3% and the highest Hit-class F1-score at 86.0% among the three tested models.
- Decision Tree achieved 73.3% accuracy and a Hit-class F1-score of 83.5%, while also showing the strongest balance in the notebook's final comparison based on macro F1 and Flop recall.
- Logistic Regression achieved 69.1% accuracy after SMOTE and showed weaker overall performance than the tree-based models.
- The model struggled considerably more with identifying Flop movies than Hit movies, reflecting the class imbalance and the difficulty of separating the negative class.

Confusion-matrix observations

Model	True Flops	False Hit predictions	Missed Hits	True Hits
Logistic Regression	15	20	53	148
Decision Tree	14	21	42	159
Random Forest	8	27	29	172

The results show that Random Forest captured the largest number of actual Hit movies, but its ability to correctly identify Flops remained limited. The final notebook notes Decision Tree as the most balanced of the three models, while Random Forest led on overall accuracy and positive-class F1.

6. Interactive Power BI Dashboard

The final Power BI report converts the analytical results into an interactive movie analytics view. The supplied dashboard screenshot shows a 1920–2026 year filter, a language slicer, KPI cards, genre/blockbuster analysis, a movie-frequency trend, an awarded-versus-not-awarded rating visual, a budget-versus-revenue scatter plot, and an interactive decomposition-style visual.

Dashboard KPIs shown

KPI	Displayed value
Total Movies	1,178
Average Budget	\$34.5M
Average Revenue	\$164.7M
Blockbuster %	9.17%
Hit %	85.14%

The dashboard's total of 1,178 reflects the financially complete subset used for the financial analytics. The dashboard defines a Blockbuster as a movie with revenue above \$500 million, as specified for this project.

Dashboard visuals and story

- Top 5 Genres by Blockbuster % highlights Adventure (39.5%), Family (21.7%), Music (14.3%), Western (6.3%), and Horror (2.9%) in the displayed breakdown.
- The Movies Frequency Trend shows the collected movie frequency increasing toward recent years, with a visible upward trend and substantial year-to-year variation.
- The Blockbuster Movies Breakdown provides hierarchical exploration by age rating and primary genre.
- The Budget vs Revenue scatter plot visually demonstrates the positive relationship between production budget and revenue and distinguishes Hit observations.
- The Awarded vs Not Awarded visual presents the average rating comparison around the overall 77.23% rating level.

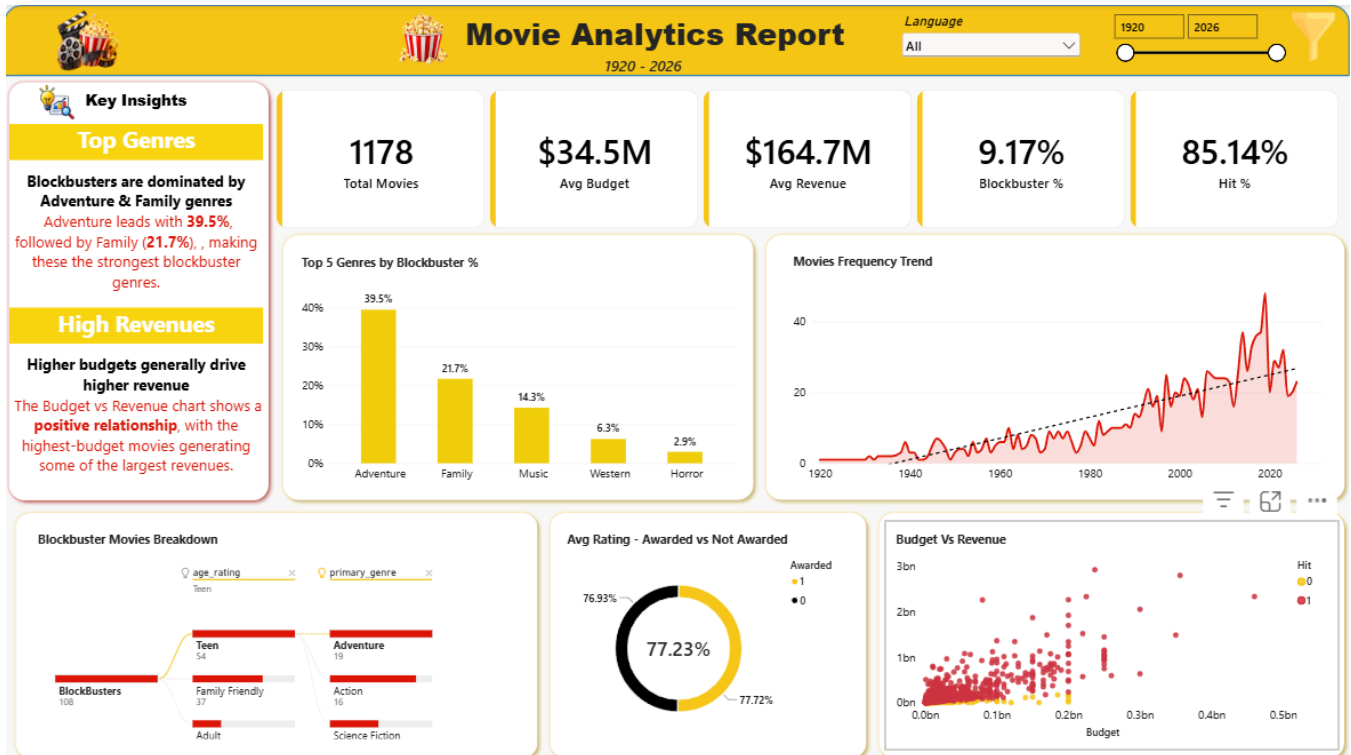


Figure 1. Final Movies Analytics Report dashboard supplied with the project.

7. Key Business Insights

The analysis supports several practical conclusions about the movie portfolio represented in the dataset.

1. Blockbuster performance is concentrated. Only 9.17% of the financially complete dashboard dataset qualifies as a Blockbuster under the project's \$500M revenue threshold.
2. Adventure is the leading blockbuster genre in the dashboard's displayed top five comparison, accounting for 39.5% of blockbuster observations in that breakdown.
3. Budget and revenue move together strongly. The approximately 0.77 correlation indicates that higher-budget movies in the sample generally tend to generate higher revenue.
4. Rating is not a strong standalone commercial predictor. The rating-revenue correlation is only about 0.14, and the EDA shows movies with moderate ratings can still achieve exceptionally high revenues.
5. Profitability is highly uneven. The very large spread between median and maximum profit shows that a small number of movies can have a disproportionate effect on aggregate financial results.
6. The financial dataset is incomplete. Only 1,178 of 2,000 movies have both budget and revenue, so profitability-related conclusions should be interpreted for the financially complete subset rather than all scraped movies.
7. The classification models are substantially better at identifying Hits than Flops, so model output should be used as a decision-support signal rather than a definitive success guarantee.

8. Recommendations

1. Use budget strategically. Since budget and revenue have a strong positive relationship in the collected data, production investment should be evaluated alongside expected market potential rather than treated independently.

2. Prioritize genre-level analysis. Adventure and other commercially strong genres should be examined further when evaluating blockbuster-oriented opportunities, while avoiding decisions based only on genre frequency.
3. Do not use rating as the sole success criterion. Ratings are useful for audience-quality analysis, but the weak rating-revenue relationship suggests that commercial performance depends on more than perceived quality.
4. Separate 'Hit' and 'Blockbuster' decisions. The project definitions capture different levels of success: IsHit is profitability-based, while Blockbuster is based on the \$500M revenue threshold.
5. Use the ML model as a supporting tool. Because Flop detection remains difficult, predictions should be combined with business context and additional pre-release variables if the model is extended.

9. Limitations

- The dataset represents the movies collected from the selected TMDb top-rated pages and should not automatically be interpreted as a complete population of all movies.
- The dataset has major class imbalance for IsHit feature. Most of the top-rated movies are Hit already.
- Financial fields are incomplete: 1,178 of the 2,000 records contain both budget and revenue.
- The IsHit target is derived from profit, so it represents the project's profitability definition rather than an externally defined industry standard.
- The Blockbuster threshold of \$500M revenue is a project-defined classification.
- The classification evaluation shows strong performance for the Hit class but relatively weak recognition of Flop movies.

10. Conclusion

The Movies Analytics project demonstrates a complete data science workflow from web data acquisition to business communication. The project collected 2,000 movie records through a two-layer Requests and BeautifulSoup scraping process, transformed the raw information into a structured dataset, and used EDA to uncover strong financial relationships, genre patterns, rating behavior, and significant outliers.

The machine-learning stage evaluated three classification approaches for the profit-based IsHit target. Random Forest achieved the highest accuracy and positive-class F1 among the tested models, while the Decision Tree provided the most balanced performance according to the final notebook comparison.

The Power BI dashboard then translated these findings into an interactive reporting experience focused on movie volume, budgets, revenue, blockbuster performance, hit rate, genre patterns, time trends, and financial relationships. Together, the analytical and dashboard outputs provide a decision-support view of movie performance within the collected dataset.

Video Link -

<https://drive.google.com/drive/folders/10zwpYn5CmlPCQojemqGo24dndfywr753?usp=sharing>

End of Report